

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

REVISED MONITORING AND REPORTING PROGRAM 5-01-153
FOR
CITY OF HANFORD
WASTEWATER TREATMENT FACILITY
KINGS COUNTY

This Monitoring and Reporting Program (MRP) is required pursuant to Water Code section 13267.

The Discharger shall not implement any changes to this MRP unless and until the Central Valley Water Board adopts or the Executive Officer issues a revised MRP. Changes to sample location shall be established with concurrence of Central Valley Water Board staff, and a description of the revised stations shall be submitted for approval by the Executive Officer. All samples shall be representative of the volume and nature of the discharge or matrix of material sampled. All analyses shall be performed in accordance with ***Standard Provisions and Reporting Requirements for Waste Discharge Requirements***, dated 1 March 1991 (Standard Provisions).

Field test instruments (such as pH) may be used provided that the operator is trained in the proper use of the instrument and each instrument is serviced and/or calibrated at the recommended frequency by the manufacturer and in accordance with manufacturer instructions.

Analytical procedures shall comply with the methods and holding times specified in the following: *Methods for Organic Chemical Analysis of Municipal and Industrial Wastewater* (EPA); *Test Methods for Evaluating Solid Waste* (EPA); *Methods for Chemical Analysis of Water and Wastes* (EPA); *Methods for Determination of Inorganic Substances in Environmental Samples* (EPA); *Standard Methods for the Examination of Water and Wastewater* (APHA/AWWA/WEF); and *Soil, Plant and Water Reference Methods for the Western Region* (WREP 125). Approved editions shall be those that are approved for use by the United States Environmental Protection Agency or the State Water Board's Division of Drinking Water Environmental Laboratory Accreditation Program. The Discharger may propose alternative methods for approval by the Executive Officer.

If monitoring consistently shows no significant variation in magnitude of a constituent concentration or parameter after at least 12 months of monitoring, the Discharger may request the MRP be revised to reduce monitoring frequency. The proposal must include adequate technical justification for the requested reduction in monitoring frequency.

A glossary of terms used within this MRP is included on page 11.

The Discharger shall monitor the following locations to demonstrate compliance with the requirements of this Order.

Monitoring Location Name	Monitoring Location Description
INF-001	Location where a representative sample of the WWTF influent can be obtain prior to any additives, treatment process, and WWTF return flow.
EFF-001	Location where a representative sample of the WWTF effluent can be obtained prior to discharge to the storage ponds.
PND-001 through PND-008	Effluent storage ponds No.1 through 8
HSW-001 through HSW-019	Hanford Supply Wells (HSW) No. 1 through 19
SLD-001	Location where a representative sample of the WWTF sludge/biosolids can be obtained after removal form the sludge drying beds.
MW-4 and MW-6 through MW-9	Monitoring wells MW-4 and MW-6 through MW-9

INFLUENT MONITORING

The Discharger shall collect influent samples at the headworks of the wastewater treatment facility (INF-001) prior to any treatment of waste. Time of a grab sample shall be recorded. Influent monitoring shall include at least the following:

<u>Frequency</u>	<u>Constituent/Parameter</u>	<u>Units</u>	<u>Sample Type</u>
Continuous	Flow	mgd	Metered
Monthly	Biochemical Oxygen Demand (BOD)	mg/L	24-hr Composite
Monthly	Total Suspended Solids (TSS)	mg/L	24-hr Composite
Monthly	Settleable Solids	mL/L	Grab

EFFLUENT MONITORING

The Discharger shall collect effluent samples at a point in the system following treatment and before discharge to the equalization basin (EFF-001). Effluent samples shall be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

<u>Frequency</u>	<u>Constituent/Parameter</u>	<u>Units</u>	<u>Sample Type</u>
Weekly	Color	unitless	Observation
Weekly	Settleable Solids	mL/L	Grab
Weekly	pH	pH Units	Grab

<u>Frequency</u>	<u>Constituent/Parameter</u>	<u>Units</u>	<u>Sample Type</u>
Daily	Total Coliform Organisms	MPN ¹ /100mL	Grab
Weekly	BOD	mg/L	Grab
Weekly	TSS	mg/L	Grab
Weekly	Electrical Conductivity	umhos/cm	Grab
Monthly	Total Dissolved Solids ³	mg/L	Grab
Monthly	Fixed Dissolved Solids	mg/L	Grab
2/Monthly ²	Ammonia as Nitrogen	mg/L	Grab
2/Monthly	Total Kjeldahl Nitrogen	mg/L	24-hr Composite
2/Monthly	Nitrate as Nitrogen	mg/L	24-hr Composite
2/Monthly	Total Nitrogen	mg/L	Calculated
Monthly	Sodium Adsorption Ratio	unitless	Calculated
2/year ⁴	General Minerals ^{5,6}	mg/L	Grab

¹ Most Probable Number

² Two times per month

³ Concurrent with EC Sampling

⁴ First and third quarter of the calendar year.

⁵ With the exception of wastewater samples, samples must be filtered. If field filtering is not feasible, samples shall be collected in unpreserved containers and submitted to the laboratory within 24 hours with a request (on the chain-of-custody form) to immediately filter then preserve the sample.

⁶ See glossary on page 11 for list of general mineral constituents.

POND MONITORING

A permanent marker (e.g., staff gages) shall be placed in the storage pond with calibration indicating the water level at design capacity and available operational freeboard. The freeboard shall be monitored on all ponds to the nearest tenth of a foot. Storage pond monitoring shall include at least the following:

<u>Frequency</u>	<u>Constituent/Parameter</u>	<u>Units</u>	<u>Sample Type</u>
Weekly	Freeboard ¹	Feet	Observation
As Required ²	DO	mg/L	Grab ³

¹ To the nearest tenth of a foot.

² If offensive odor is detected by or brought to the attention of WWTF personnel, monitor affected pond(s) daily until dissolved oxygen >1.0 mg/L.

³ DO shall be measured between 8:00 am and 9:00 am and shall be taken opposite the pond inlet at a depth of approximately one-foot.

The Discharger shall inspect the condition of the storage pond weekly and record visual observations in a bound logbook. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation,

scum, or debris are accumulating on the storage pond surface and their location; whether burrowing animals or insects are present; and the color of the reservoirs (e.g., dark green, dull green, yellow, gray, tan, brown, etc.). A summary of the entries made in the log during each month shall be submitted along with the monitoring report the following month. If the Discharger finds itself in violation of Discharge Specifications B.10, B.11, B.12, or B.15, the Discharger shall briefly explain the action taken or to be taken to correct the violation. The Discharger shall certify in each November monitoring report that it is in compliance with Discharge Specification B.18.

SOURCE WATER MONITORING

The Discharger shall establish source sample stations where representative samples of the City of Hanford's water supply can be obtained. The results shall be reported as flow weighted average and be supplemented with supporting calculations. Source water monitoring shall include:

<u>Frequency</u>	<u>Constituent/Parameter</u>	<u>Units</u>	<u>Sample Type</u>
Quarterly ¹	EC	umhos/cm	Grab
Once every 3 years ²	TDS	mg/L	Grab

¹ January, April, July and October

² Coincident with monitoring requirements by Division of Drinking Water.

PRETREATMENT MONITORING

The Discharger shall submit annually a report to the Regional Board, with copies to EPA Region 9 and the State Board, describing the Discharger's pretreatment activities over the previous 12 months. In the event that the Discharger is not in compliance with any conditions or requirements of this Order, including noncompliance with pretreatment audit/ compliance inspection requirements, the Discharger shall also include reasons for noncompliance and stat how and when the Discharger shall comply with such conditions and requirements. The annual pretreatment report is due by **28 February of each year**. In addition to the information required in the annual pretreatment report, the Discharger shall submit a quarterly report **by the 1st day of the second month following the end of each calendar quarter**. The report shall contain, but not limited to, the items in Standard Provision E.7.

If none of the items in Standard Provision E.7 exist, at a minimum a letter indicating that all industries are in compliance and no violations or changes to the pretreatment program have occurred during the quarter must be submitted. The information required in the fourth quarter report shall be included as part of the annual report.

Signed copies of the report shall also be submitted the EPA Regional Administrator and the State Board at the following addresses, or a advised in writing subsequent to adoption of this Order:

Regional Administrator
U.S. Environmental Protection Agency W-5
75 Hawthorn Street
San Francisco, CA 94105

State Water Resources Control Board
Division of Water Quality
P.O. Box 100
Sacramento, CA 95812

SLUDGE MONITORING

To monitor whether discharges to the WWTF are interfering with the treatment process or lessening biosolids quality, the Discharger shall collect a composite sample of sludge at least annually in accordance with EPA's POTW SLUDGE SAMPLING AND ANALYSIS GUIDANCE DOCUMENT, AUGUST 1989, and test for the following metals at SLD-001:

Arsenic	Copper	Nickel
Cadmium	Lead	Selenium
Molybdenum	Mercury	Zinc

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report. Prior to any disposal or land application of sewage sludge, or removal of sewage sludge from the WWTF, the monitoring and record keeping requirements of 40 CFR 503 shall be met.

GROUNDWATER MONITORING

Prior to collecting samples, each monitoring well shall be adequately purged to remove water that has been standing within the well screen and casing that may not be chemically representative of formation water. Depending on the hydraulic conductivity of the geologic setting, the volume removed during purging is typically 3 to 5 volumes of the standing water within the well casing and screen, or additionally the filter pack pore volume.

At least quarterly and concurrently with groundwater quality sampling, the Discharger shall measure the water level in each well. The Discharger shall report groundwater level data as groundwater depth (in feet and hundredths) as a groundwater surface elevation (in feet and hundredths above mean sea level). The horizontal geodetic location for each monitoring well shall be provided where the point of beginning shall be described by the California State Plane Coordinate System, 1983 datum.

The Discharger shall monitor monitoring groundwater wells and any subsequent additional wells, for the following:

<u>Frequency</u>	<u>Constituent/Parameter</u>	<u>Units</u>	<u>Sample Type</u>
Quarterly	EC	umhos/cm	Grab
Quarterly	TDS	mg/L	Grab
Quarterly	Total Coliform Organism	MPN/100mL	Grab
Quarterly	Total Organic Carbon	mg/L	Grab
Quarterly	Ammonia as Nitrogen	mg/L	Grab
Quarterly	Nitrate as Nitrogen	mg/L	Grab
Quarterly	Total Kjeldahl Nitrogen	mg/L	Grab
Quarterly	Total Nitrogen	mg/L	Calculated
Quarterly	General Minerals ^{1,2}	mg/L	Grab

¹ With the exception of wastewater samples, samples must be filtered. If field filtering is not feasible, samples shall be collected in unpreserved containers and submitted to the laboratory within 24 hours with a request (on the chain-of-custody form) to immediately filter then preserve the sample.

² See glossary on page 11 for list of general mineral constituents.

After one full year of groundwater monitoring, the Discharger shall analyze monitoring data from background well(s) to compute background water quality values for each monitored constituent and to perform an initial assessment of whether there is evidence of impact from the discharge. To complete this task, the Discharger shall use monitoring data from background and boundary wells in an appropriate data analysis method as described in Title 27, section 20415(e)(7-9) (hereafter Data Analysis Method). Reports thereafter shall be submitted quarterly by the **1st day of the second month** after the prescribed sample collection and shall include the same analysis. The Discharger shall perform the Data Analysis Method on the following constituents.

Groundwater Constituents to Evaluate Using Data Analysis Method

Alkalinity (as CaCO ₃)	Phosphorus, Total Dissolved (P)
Ammonia nitrogen (as NH ₃ -N)	Potassium
Calcium	Sodium
Chloride	Sulfate
Hardness (as CaCO ₃)	Total Dissolved Solids (TDS)
Iron	Total Kjeldahl Nitrogen (TKN)
Magnesium	Total Nitrogen
Nitrate nitrogen (as NO ₃ -N)	Total Organic Carbon (TOC)

If the Discharger during any quarterly data evaluation finds statistically significant evidence of an increase at boundary wells compared to background levels of TKN or TOC, or evidence of exceedances of Groundwater Limitation D.1, the Discharger shall conclude that it is in violation of waste discharge requirements unless it can demonstrate an offsite source. The

Discharger shall describe the data analysis method used as well as the criteria it used for determining “statistically significant evidence,” and submit within two weeks, of confirmation, a written report pursuant to Standard Provision B.1.

REPORTING

All monitoring results shall be reported in **Quarterly Monitoring Reports** which are due by the first day of the second month after the calendar quarter. Therefore, monitoring reports are due as follows:

First Quarter Monitoring Report:	1 May
Second Quarter Monitoring Report:	1 August
Third Quarter Monitoring Report:	1 November
Fourth Quarter Monitoring Report:	1 February

The Central Valley Water Board has gone to a Paperless Office System. All regulatory documents, submissions, materials, data, monitoring reports, and correspondence shall be converted to a searchable Portable Document Format (PDF) and submitted electronically. Documents that are less than 50MB should be mailed to: centralvalleyfresno@waterboards.ca.gov. Documents that are 50MB or larger should be transferred to a disc and mailed to the appropriate regional water board office, in this case 1685 E Street, Fresno, CA, 93706.

To ensure that your submittals are routed to the appropriate staff, the following information block should be included in any email used to transmit documents to this office:

Program: Non-15, WDID: 5D160103001, Facility Name: City of Hanford Wastewater Treatment Facility, Order: 5-01-153

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly, whether the Discharger complies with waste discharge requirements, and shall discuss any violations that occurred during the reporting period and all actions taken or planned for correcting violations, such as operation or facility modifications. If the Discharger has previously submitted a report describing corrective actions or a time schedule for implementing the corrective actions, reference to the previous correspondence is satisfactory.

In addition to the details specified in Standard Provision C.3, monitoring information shall include the method detection limit (MDL) and the reporting limit (RL) or practical quantitation limit (PQL). If the regulatory limit for a given constituent is less than the RL (or PQL), then

any analytical results for that constituent that are below the RL (or PQL) but above the MDL shall be reported and flagged as estimated.

Laboratory analysis reports do not need to be included in the monitoring reports; however, the laboratory reports must be retained for a minimum of three years in accordance with Standard Provision C.3.

All monitoring reports shall comply with the signatory requirements in Standard Provision B.3. Monitoring data or discussions submitted concerning WWTF performance must also be signed and certified by the chief plant operator. If the chief plant operator is not in direct line of supervision of the laboratory function for a Discharger conducting any of its own analyses, reports must also be signed and certified by the chief of the laboratory.

All monitoring reports that involve planning, investigation, evaluation, or design, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code sections 6735, 7835, and 7835.1.

In the future, the State or Central Valley Water Board may notify the District to electronically submit and upload monitoring reports using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site <http://www.waterboards.ca.gov/ciwqs/index.html> or similar system. Electronic submittal to CIWQS, when implemented, will meet the requirements of our Paperless Office System.

A. All Quarterly Monitoring Reports shall include the following:

Wastewater Reporting

1. The results of Influent, Effluent, and Pond Monitoring specified on pages 2 through 4.
2. For each month of the quarter, calculation of the maximum daily flow and the monthly average flow.
3. For each of the quarters, calculation of the 12-month rolling average EC of the discharge using the EC value for that month averaged with EC values for the previous 11 months.
4. For each month of the quarter, calculation of the monthly average effluent BOD₅ and TSS concentrations, and calculation of the percent removal of BOD₅ and TSS compared to the influent.

5. A summary of the notations made in the pond monitoring log during each quarter. Copies of log pages covering the quarterly reporting period shall not be submitted unless requested by Central Valley Water Board staff.

Groundwater Reporting

1. The results for Groundwater Monitoring specified on pages 5 through 7.
2. For each monitoring well, a table showing constituent concentration for at least five previous years, if available, through the current quarter.
3. A groundwater contour map based on groundwater elevations for that quarter. The map shall show the gradient and direction of groundwater flow under/around the facility and/or effluent disposal area(s). The map shall also include the location of monitoring wells and wastewater discharge areas.

Source Water Reporting

1. The results of Source Water Monitoring specified on page 4.
2. For each month of the quarter, calculation of the flow-weighted 12-month rolling average EC of the source water using monthly flow data and the source water EC values for the most recent four quarters.

B. Fourth Quarter Monitoring Reports, in addition to the above, shall include the following:

Wastewater Treatment Facility Information

1. The names, certificate grades, and general responsibilities of all persons in charge of wastewater treatment and disposal.
2. The names and telephone numbers of persons to contact regarding the WWTF for emergency and routine situations.
3. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibrations (Standard Provision C.4).
4. A statement whether the current operation and maintenance manual, sampling plan, and contingency plan, reflect the WWTF as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.

5. The results of an annual evaluation conducted pursuant to Standard Provision E.4 and a figure depicting monthly average discharge flow for the previous five calendar years.
6. A summary and discussion of the compliance record for the reporting period. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with this Order.

Sludge/Biosolids Monitoring

1. Annual production totals in dry tons or cubic yards.
2. A description of disposal methods, including the following information related to the disposal methods used. If more than one method is used, include the percentage disposed of by each method.
 - a. For landfill disposal, include: the name and location of the landfill, and the Order number of WDRs that regulate it.
 - b. For land application, include: the location of the site, and the Order number of any WDRs that regulate it.
 - c. For incineration, include: the name and location of the site where incineration occurs, the Order number of WDRs that regulate the site, the disposal method of ash, and the name and location of the facility receiving ash (if applicable).
 - d. For composting, include: the location of the site, and the Order number of any WDRs that regulate it.

Pretreatment Monitoring

1. Annual pretreatment report specified on pages 4 and 5.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

Ordered by:

Clay L. Rodgers

for PAMELA C. CREEDON, Executive Officer

10/17/2016

(Date)

GLOSSARY

BOD ₅	Five-day biochemical oxygen demand		
CBOD	Carbonaceous BOD		
DO	Dissolved oxygen		
EC	Electrical conductivity at 25° C		
FDS	Fixed dissolved solids		
NTU	Nephelometric turbidity unit		
TKN	Total Kjeldahl nitrogen		
TDS	Total dissolved solids		
TSS	Total suspended solids		
Continuous	The specified parameter shall be measured by a meter continuously.		
24-Hour Composite	Samples shall be a flow-proportioned composite consisting of at least eight aliquots.		
Daily	Samples shall be collected at least every day.		
Twice Weekly	Samples shall be collected at least twice per week on non-consecutive days.		
Weekly	Samples shall be collected at least once per week.		
Twice Monthly	Samples shall be collected at least twice per month during non-consecutive weeks.		
Monthly	Samples shall be collected at least once per month.		
Bimonthly	Samples shall be collected at least once every two months (i.e., six times per year) during non-consecutive months.		
Quarterly	Samples shall be collected at least once per calendar quarter. Unless otherwise specified or approved, samples shall be collected in January, April, July, and October.		
Semiannually	Samples shall be collected at least once every six months (i.e., two times per year). Unless otherwise specified or approved, samples shall be collected in April and October.		
Annually	Samples shall be collected at least once per year. Unless otherwise specified or approved, samples shall be collected in October.		
mg/L	Milligrams per liter		
mL/L	milliliters [of solids] per liter		
ug/L	Micrograms per liter		
umhos/cm	Micromhos per centimeter		
mgd	Million gallons per day		
MPN/100 mL	Most probable number [of organisms] per 100 milliliters		
General Minerals	Analysis for General Minerals shall include at least the following:		
	Bicarbonate	Hardness	Potassium
	Boron	Iron	Sodium
	Calcium	Magnesium	Sulfate
	Carbonate	Manganese	
	Chloride	Phosphate	
	General Minerals analyses shall be accompanied by documentation of cation/anion balance.		